DUNMAN SECONDARY SCHOOL

CANDIDATE NAME	
CLASS	
INDEX NUMBER	

PRELIMINARY EXAMINATION 2023
SECONDARY 4 EXPRESS

MATHEMATICS **4052/02**

Paper 2

24 August 2023

2 hours 15 minutes

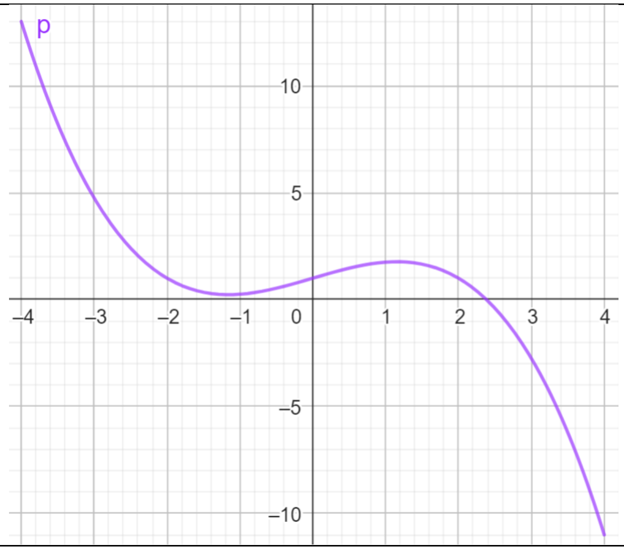
Marking Scheme

(Updated 1 Dec 2022 by HOD/Math aligned to Marking Scheme for Specimen Papers 4049)

Question	Answer		
1a	$7x+8y=14$ Eqn 1		
	$x+4y=12$ Eqn 2		
	Eqn 2 × 2	$2x+8y=24$	Eqn 3
	Eqn 1 – Eqn 3	$5x=-10$	
		$x=-2$	
	Sub $x=-2$ into Eqn 2	$y = 3.5$	
1b	$\frac{8x^2}{27y^3} \div \frac{4}{3x^3y} = \frac{8x^2}{27y^3} \times \frac{3x^3y}{4}$		
	$= \frac{2x^5}{9y^2}$		
1c	$\frac{2x^2-18}{x^2-6x+9} = \frac{2(x+3)(x-3)}{(x-3)^2}$		
	$= \frac{2(x+3)}{x-3}$		

Question	Answer
2a	$\angle ABC = \angle BCD$ (interior angles of regular polygon)
	$\angle XBC = 180° - \angle ABC$ (adj. angles on a str. line)
	$\angle XCB = 180° - \angle BCD$ (adj. angles on a str. line)
	Since $\angle XBC = \angle XCB$, triangle XBC is an isosceles triangle
2bi	External angle = 24°
	$n = \frac{360}{24} = 15$
2bii	132°
2biii	12°
2biv	Angle ABD = 156° – 12° = 144°
	Angle ABD = 180° – 144° (Angles in opp. segment)
	= 36°

Question	Answer
3a	221
3b	$(2n+3)(4n-3)$
	$= 8n^2+6n-9$
3c	Because the terms are products of odd numbers.
3d	$[8(p+1)^2+6(p+1)-9]-[8p^2+6p-9]$
	$= [8(p^2+2p+1)+6p+6-9]-8p^2-6p+9]$
	$= 8p^2+16p+8+6p+6-9-8p^2-6p+9$
	$= 16p+14$
3e	Because the difference, $2(8p+7)$, is a multiple of 2.
	<u>OR</u> Because each term in the sequence is odd. Hence their difference will be even,

Question	Answer
4a	13
4b	
4c	Well-balanced tangent line drawn
	–5.7
4di	$a = 2$
	$b = -1$
4dii	Proper straight line drawn
	$x = 1.35$
4e	0.2 and 1.8

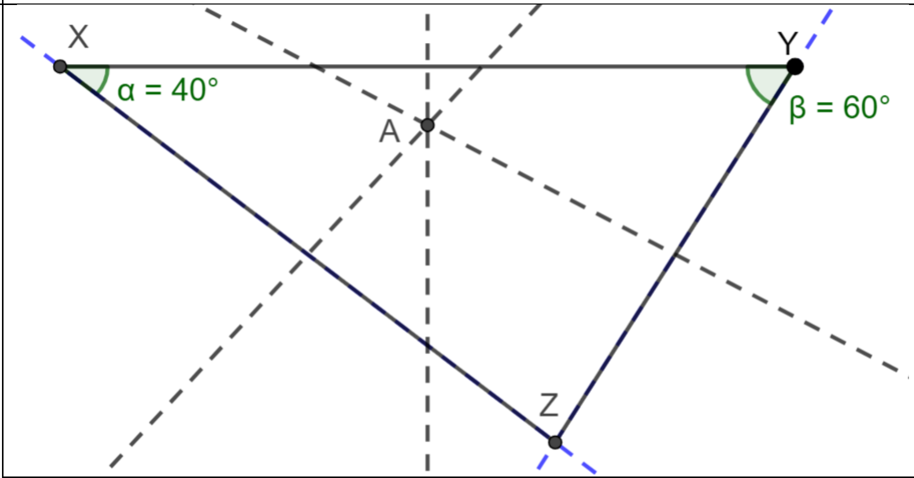
Question	Answer
5a	$\dfrac{15}{x}$
5b	$\dfrac{15}{x+1}$
5c	$\dfrac{15}{x}+\dfrac{15}{x+1}=\dfrac{960}{60}$
	$\dfrac{15(x+1)}{x(x+1)}+\dfrac{15x}{(x+1)x}=16$
	$15x+15+15x=16(x^2+x)$
	$0=16x^2+16x-30x-15$
	$16x^2-14x-15=0$
5d	$x=\dfrac{-(-14)\pm\sqrt{(-14)^2-4(16)(-15)}}{2(16)}$
	$x=1.5$ or -0.625
5e	Rate = $\dfrac{15}{1.5+1}$
	= 6 dumpling per minute
	In 1 hour, 6×60
	= 360

Question	Answer
6a	$QR = \sqrt{19^2 + 17^2 - 2(19)(17)\cos 41°}$
	$= 12.745 = 12.7$ m
6b	$\dfrac{\sin \angle PRQ}{19} = \dfrac{\sin 41°}{12.745}$
	$\angle PRQ = 77.970° = 78.0°$
6c	$\tan \angle PQT = \dfrac{5.2}{19}$
	$\angle PQT = 15.3\ °$
6d	The point with the largest angle of depression will be the point with shortest distance.
	Let the shortest distance be d
	$\dfrac{d}{17} = \sin 41°$
	$d = 11.2$ m

Question	Answer
7ai	35%
7aii	1.85
7aiii	2.1 – 1.2
	= 0.9
7aiv	He should buy from Company *A* because
	the bolts in Company *A* are lighter because Company *A*’s median of 1.85 g is smaller than Company *B*’s median of 1.9 g.
	the spread in Company *A*’s bolts is smaller because Company *A*’s interquartile range of 0.9 g is smaller than Company *B*’s interquartile range of 1 g.
7bi	$\frac{1}{2}\times\frac{4}{3}\times\pi\times0.5^3+\pi\times0.3^2\times1.3$
	= 0.629 cm^3
7bii	$\frac{1}{2}\times4\times\pi\times0.5^2+2\times\pi\times0.3\times1.3+\pi\times0.5^2$
	= 4.81 cm^2

Question	Answer
8a	Since *AC* passes through centre and bisects BD, $\angle AHD=90°$. (perpendicular bisector of chord *BD*)
	$\angle EAH=90°$ (tangent perpendicular to radius)
	Since $\angle AHD+\angle EAH=180°$, *AE* is parallel to *BD*.
8b	$\angle AFE=\angle DFB$ (vert. opp. angles)
	$\angle FAE=\angle FDB$ (alt. angles, *AE* is parallel to *BD*)
	$\angle FEA=\angle FBD$ (alt. angles, *AE* is parallel to *BD*)
	Triangle *AFE* is similar to triangle *DFB*. (AA similarity test)
8ci	$\overrightarrow{FD}=2\mathbf{q}$
	$\overrightarrow{DH}=-\mathbf{p}$
	$\overrightarrow{AH}=-\mathbf{p}+3\mathbf{q}$
8cii	Triangle *AGE* is congruent to triangle *HGB*.

	$\overrightarrow{AG} = \frac{1}{2}(-\mathbf{p} + 3\mathbf{q})$
	$= -\frac{1}{2}\mathbf{p} + \frac{3}{2}\mathbf{q}$
8d	$\overrightarrow{EF} = -\mathbf{p} + \mathbf{q}$
	$\overrightarrow{FG} = -\mathbf{q} + -\frac{1}{2}\mathbf{p} + \frac{3}{2}\mathbf{q}$
	$= -\frac{1}{2}\mathbf{p} + \frac{1}{2}\mathbf{q}$
	$= \frac{1}{2}\overrightarrow{EF}$
	Area of triangle AFE : area of the triangle $AFG = 2 : 1$

Question	Answer
9a	mean = 200.4
	standard deviation = 21.1
9b	slowest possible speed is 158.2 km/h
	Longest possible time = $\dfrac{135}{158.2}$
	= 0.853 h
9c	
	Shortest distance to each town = 68.5 km
	Fastest speed = 242.6 km/h
	Shortest time to each town = $\dfrac{68.5}{242.6}$ = 0.282 h = 16.92 min
	No, Brian is wrong.